

Git Tutorial: Branches and Pull Requests

COMS W4156 (Fall 2024)

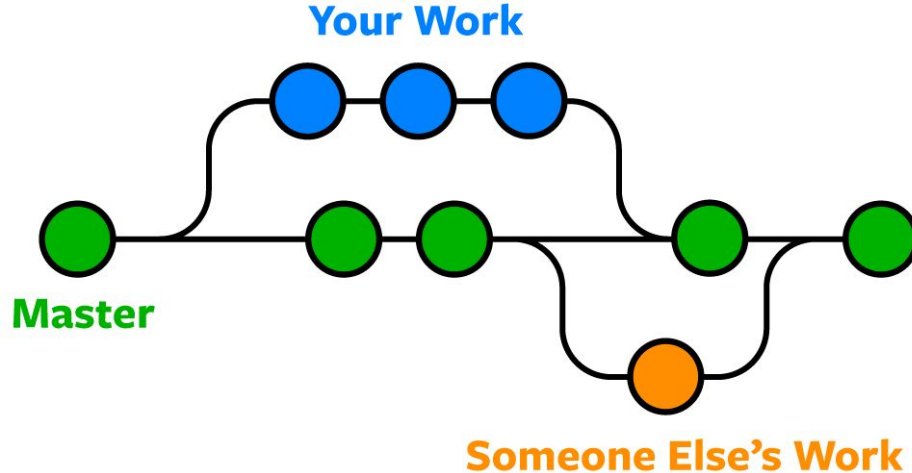
Prerequisites

- Knowledge of version control systems and basic git workflow
 - See Prof. Jae's Git Tutorial (<https://cs3157.github.io/www/2024-1/guides/git-tutorial.pdf>)
- Github account

Disclaimer: This tutorial provides a focused overview, highlighting only the essential information and commands necessary for completing your projects. It is not intended to be a comprehensive overview.

Git Branches

- **Definition:** A lightweight movable pointer to a commit.
- **Purpose:** Allows multiple lines of development, enabling parallel work.
- **Use Case:** Feature development, bug fixes, experiments, etc.



Key Commands for Branching

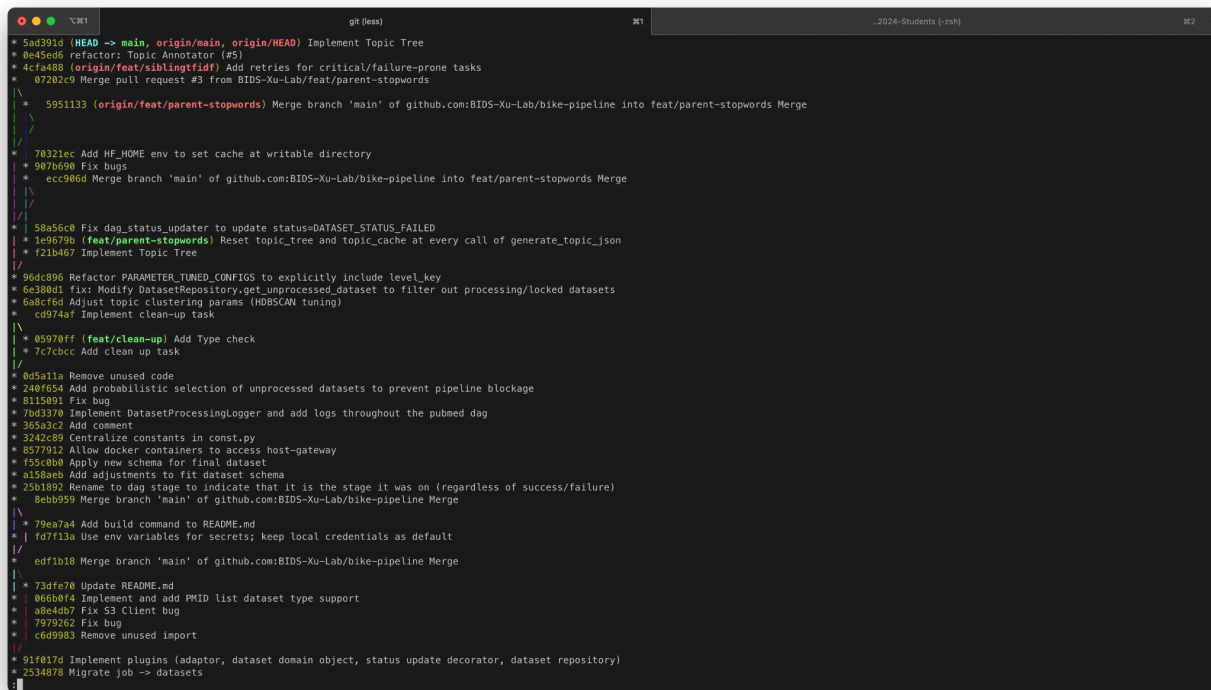
- **Create a New Branch:** `git branch <branch-name>`
- **Switch Branches:** `git checkout <branch-name>` **or** `git switch <branch-name>`
- **Create and Switch:** `git checkout -b <branch-name>` **or** `git switch -c <branch-name>`
- **List Branches:** `git branch`

```
> git branch feat/first-fix  
> git checkout feat/first-fix  
Switched to branch 'feat/first-fix'
```

```
> git checkout -b feat/graceful-shutdown  
Switched to a new branch 'feat/graceful-shutdown'
```

Visualizing Branches

`git log --oneline --graph --decorate`



```
git (less) M1 .2024-Students (-vsh) M2
* 5ad391d (HEAD -> main, origin/main, origin/HEAD) Implement Topic Tree
* 0e45cd6 refactor: Topic Annotator (AS)
* 4cfa488 (origin/feat/siblingtidf) Add retries for critical/failure-prone tasks
* 07202c9 Merge pull request #3 from BIDS-Xu-Lab/feat/parent-stopwords
* 5951133 (origin/feat/parent-stopwords) Merge branch 'main' of github.com:BIDS-Xu-Lab/bike-pipeline into feat/parent-stopwords Merge
* 70321ec Add HF_HOME env to set cache at writable directory
* 9070690 Fix bugs
* ecc906d Merge branch 'main' of github.com:BIDS-Xu-Lab/bike-pipeline into feat/parent-stopwords Merge
* 58a56c0 Fix dag_status_updater to update status=DATASET_STATUS_FAILED
* 1e9679b (feat/parent-stopwords) Reset topic_tree and topic_cache at every call of generate_topic_json
* f21b467 Implement Topic Tree
* 964c096 Refactor PARAMETER_TUNED_CONFIGS to explicitly include level key
* 6e380d1 fix: Modify DatasetRepository.get_unprocessed_dataset to filter out processing/locked datasets
* 6a8cf6d Adjust topic clustering params (HDBSCAN tuning)
* cd974af Implement clean-up task
* 05970ff (feat/clean-up) Add Type check
* 7c7cbcc Add clean up task
* 0d5a11a Remove unused code
* 240f054 Add probabilistic selection of unprocessed datasets to prevent pipeline blockage
* 8115091 Fix bug
* 7bd3370 Implement DatasetProcessingLogger and add logs throughout the pubmed dag
* 365a3c2 Add comment
* 3242c89 Centralize constants in const.py
* 0379912 Allow docker containers to access host-gateway
* f55c0b0 Apply new schema for final dataset
* a158aeb Add adjustments to fit dataset schema
* 25b1892 Rename to dag stage to indicate that it is the stage it was on (regardless of success/failure)
* 8ebb959 Merge branch 'main' of github.com:BIDS-Xu-Lab/bike-pipeline Merge
* 79ea7a4 Add build command to README.md
* fd7f13a Use env variables for secrets; keep local credentials as default
* edf1b18 Merge branch 'main' of github.com:BIDS-Xu-Lab/bike-pipeline Merge
* 73dfe70 Update README.md
* 066b0f4 Implement and add PMID list dataset type support
* a8e4ub7 Fix S3 Client bug
* 7970262 Fix bug
* c6d9983 Remove unused import
* 910017d Implement plugins (adaptor, dataset domain object, status update decorator, dataset repository)
* 2534878 Migrate job -> datasets
```

Pull Requests (PR)

- **Definition:** A method of submitting contributions to a repository.
- **Purpose:** Facilitates code review and discussion before merging changes.

chore: Add base implementation of test suite #1

[Open](#) binchoi-yale wants to merge 1 commit into `main` from `feat/test-demo-slide`

Conversation 0 Commits 1 Checks 0 Files changed 1 +4 -0

binchoi-yale commented 2 minutes ago

Basic test suite set-up.

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Flowchart

```
graph TD; A --> B; A --> C; B --> D; C --> D;
```

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

binchoi-yale

Labels

enhancement

Projects

None yet

Milestone

No milestone

Notifications

Unsubscribe

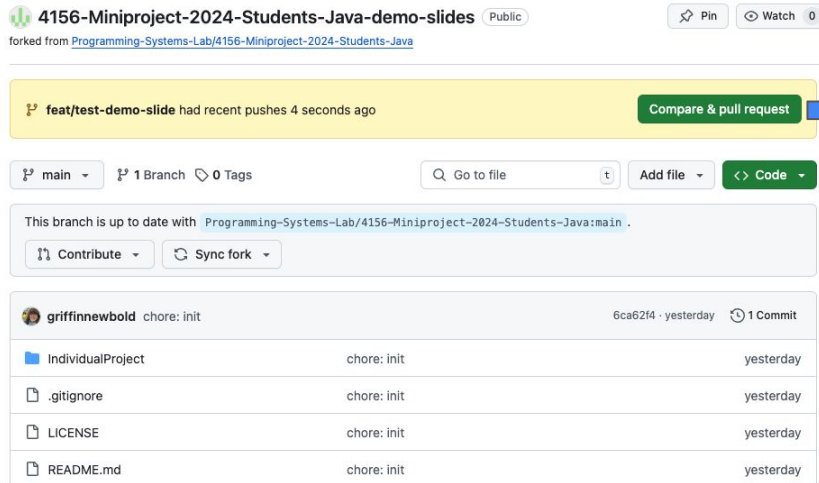
You're receiving notifications because you were assigned.

2 participants

Add test suite 56df004

Pull Requests (PR)

1. **Make relevant changes on newly created branch**
2. **Push Branch:** `git push --set-upstream origin <branch-name>`
3. **Navigate to Repository on Github**
4. **Click 'Compare & pull request' button**



4156-Miniproject-2024-Students-Java-demo-slides Public

forked from Programming-Systems-Lab/4156-Miniproject-2024-Students-Java

feat/test-demo-slide had recent pushes 4 seconds ago

Compare & pull request

main 1 Branch 0 Tags

Go to file Add file Code

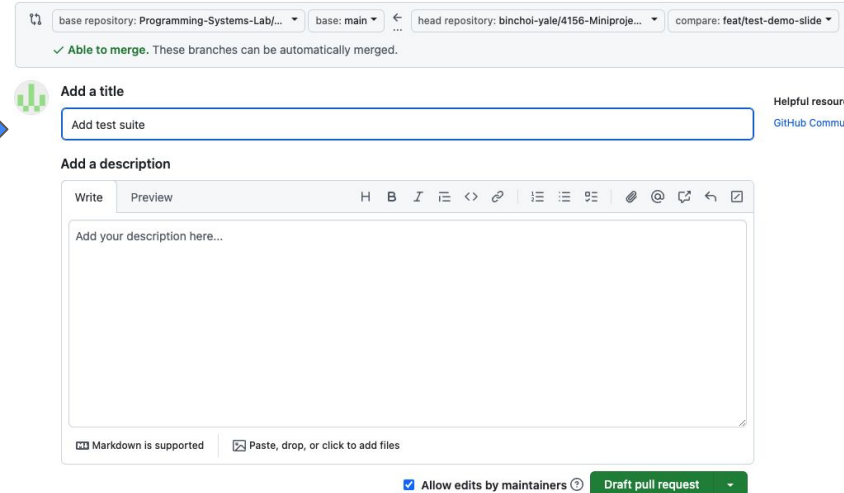
This branch is up to date with Programming-Systems-Lab/4156-Miniproject-2024-Students-Java:main

Contribute Sync fork

griffinnewbold chore: init	6ca62f4 · yesterday	1 Commit
IndividualProject	chore: init	yesterday
.gitignore	chore: init	yesterday
LICENSE	chore: init	yesterday
README.md	chore: init	yesterday

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons](#)



base repository: Programming-Systems-Lab/... base: main head repository: binchoi-yale/4156-Miniproje... compare: feat/test-demo-slide

✓ Able to merge. These branches can be automatically merged.

Add a title

Add test suite

Add a description

Write Preview

Add your description here...

Markdown is supported Paste, drop, or click to add files

Allow edits by maintainers Draft pull request

Pull Requests (PR)

5. Select appropriate base repository !!!

- By default, the original (upstream) repository will be selected: `Programming-Systems-Lab/...`
- You **must select your own forked repository** as the base repository. No credit for PRs to other repos.

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons](#)

base repository: Programming-Systems-Lab/... base: main head repository: binchoi-yale/4156-Miniproject-2024-Students compare: feat/test-demo-slide

Choose a Base Repository

binchoi-

binchoi-yale/4156-Miniproject-2024-Students

Helpful resource

Add a description

Write Preview

Add your description here...

Markdown is supported Paste, drop, or click to add files

☒ Allow edits by maintainers Draft pull request

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#)

base: main compare: feat/test-demo-slide ✓ Able to merge. These branches can be automatically merged.

Add a title

Add test suite

Add a description

Write Preview

Add your description here...

Markdown is supported Paste, drop, or click to add files

Create pull request

Pull Requests (PR)

6. Fill in PR Details (description, line comments, etc.)

7. Assign reviewers & submit PR (fyi. not applicable for individual assignments)

chore: Add base implementation of test suite #1

Edit <> Code

binchoi-yale wants to merge 1 commit into `main` from `feat/test-demo-slide`

Conversation 0

Commits 1

Checks 0

Files changed 1

+4 -0



binchoi-yale commented 2 minutes ago

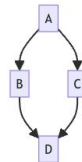
Owner

Basic test suite set-up.

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Flowchart



Reviews

No reviews

Still in progress? [Convert to draft](#)

Assignees

binchoi-yale

Labels

enhancement

Projects

None yet

Milestone

No milestone

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You're receiving notifications because you were assigned.

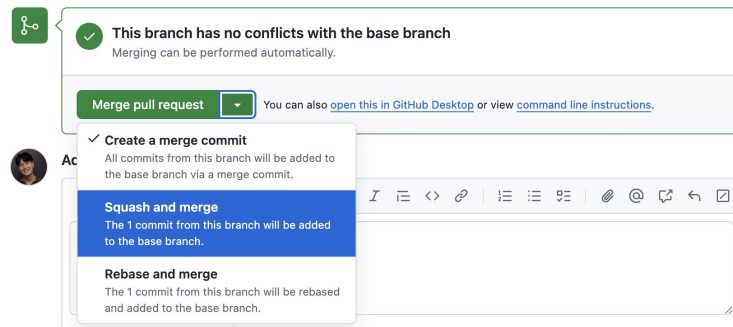
2 participants

Add test suite

56df004

Pull Requests (PR)

- **Reviewing PRs:** Comment, approve, or request changes. (n/a for individual projects)
- **Merging PRs:** Options include merge commit, rebase and merge, or squash and merge.
 - We recommend that you use **Squash and Merge**
- **Closing PRs:** Can be closed without merging if necessary.



Interacting with Pull Requests: Revert

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).

 base: main  compare: revert-2-chore/add-test ✓ Able to merge. These branches can be automatically merged.



Add a title

Revert "Add unit tests for course repo"

Add a description

Write Preview

H B I                                    

Demo

Q&A

Feel free to drop any questions you have in the comments of the Ed Discussion post #21

[Q&A] Branches and Pull Requests #21



Bin Choi **STAFF**

23 seconds ago in **Individual Miniproject**



UNPIN



STAR



WATCHING

3

VIEWS



If you have any questions or concerns about working with branches and pull requests in your mini-project, drop them in the comments below.

Demo slides can be found [here](#).

Comment Edit Delete Endorse ...



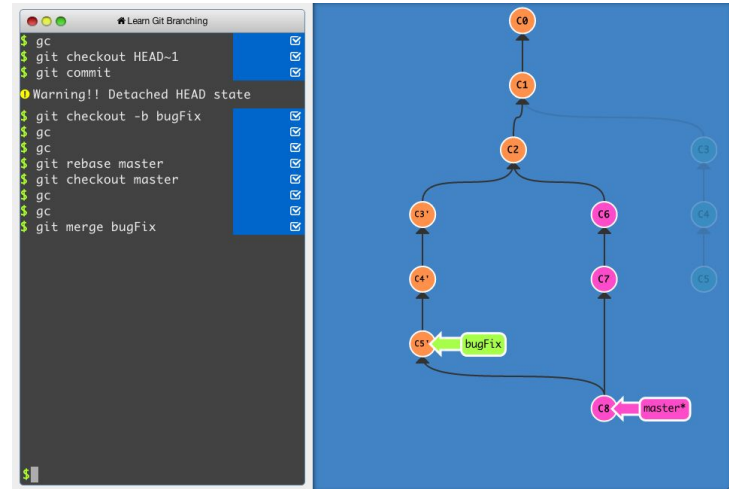
Add comment

TL;DR

- **Git Branches:** Enable parallel development by allowing multiple lines of work
- **Key Commands**
 - `git branch`
 - `git checkout`
 - `git switch`
- **Pull Requests:** Facilitate code review and merging changes
- **Remember:** Select your own forked repository as base repository for all of your PRs!
- **Tip:** Use Squash and Merge for clean commit history

Additional Resources

- <https://learngitbranching.js.org/>



- <https://cs3157.github.io/www/2024-1/guides/git-tutorial.pdf>
- <https://docs.github.com/en/get-started/using-github/github-flow>